

ROBINSON VIDVA

Computational Biologist

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SUMMARY

Computational biologist with 18+ years of experience in bioinformatics, multi-omics data analysis, neuroscience, immunology, immuno-oncology, and drug discovery. Research focuses on integrating transcriptomics, metabolomics, and genomics data to understand disease mechanisms and evaluate therapeutic interventions, from molecular pathways to behavioral outcomes, with direct applications to improving pediatric health outcomes.

Work has contributed to 7 peer-reviewed publications, 2 preprints under review, 14 conference presentations, 2 granted patents, and 145+ citations. Research spans from immune-mediated cancer therapies and autoimmune disease modeling to current work on neurodevelopmental disorders, applying RNA-seq, metabolomics, and behavioral analysis to study how preterm birth affects brain development and treatment responses.

Proficient in Python, R, SQL, and MATLAB, with expertise in specialized tools like DeepLabCut and BSOD for behavioral analysis. Experience includes managing research teams and translating complex computational findings into biological insights that advance disease understanding and therapeutic development.

KEY ACHIEVEMENTS

- **Neurodevelopmental Data Analysis**

Applied multi-omics approaches (RNA-seq, metabolomics) to study preterm birth effects on cerebellar development, analyzing 200+ behavioral videos to quantify deficits in mouse models.
- **Immuno-Oncology Research**

Analyzed cancer genomics data from 500+ cell lines and TCGA/MSK datasets to identify therapeutic targets, contributing to 7 ASH conference abstracts and 1 leukemia publication.
- **Predictive Disease Modeling**

Developed autoimmune disease pathway model involving 8 cell types, validated with in vivo data, contributing to 3 publications and 5 conference abstracts.
- **Automated Pipeline Development**

Built scalable workflows for large biological datasets, reducing analysis time by 30% and improving research reproducibility.
- **Drug Discovery Innovation**

Developed novel therapeutic combinations through computational analysis, securing 2 patents for cancer and autoimmune treatments.
- **Team Leadership**

Led teams of 10+ members and conducted 25+ training sessions in computational biology and data analytics methods.

EDUCATION

- 07/2024 - 01/2025

United States

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Master of Science (MS) - Data Analytics

Western Governors University
- 11/2012 - 02/2020

India

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Master of Business Administration (MBA) - Project Management

Sikkim Manipal University
- 07/2003 - 04/2007

India

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Bachelor of Technology (B.Tech.) - Biotechnology

Bharathidasan University

WORK EXPERIENCE

03/2025 - Present

United States

Research Technician

Children's National Hospital

<https://www.childrensnational.org>

- Perform multi-omics analyses (RNA-seq, metabolomics, genomics) on mouse/human datasets using Python, R, SQL.
- Identify DEGs, pathways, GO terms linked to neurodevelopmental deficits, NEC, SIDS.
- Apply DeepLabCut, BSOD, and custom Python scripts to analyze 3-chamber behavioral assays in mouse models, quantifying how physical therapy and environmental enrichment (EE) interventions impact neurodevelopmental outcomes in preterm birth research.
- Build ML and statistical models to analyze Erasmus Ladder motor coordination data, identifying significant gait and cerebellar deficits in preterm models. Create visualizations to highlight motor learning impairments and sensorimotor integration patterns.
- Analyze neonatal growth data using Fenton charts, performing statistical analysis and data cleaning to track developmental and neurodevelopmental trajectories in preterm infants with and without NEC. Create visualizations to communicate growth patterns and clinical findings.

10/2019 - 06/2022

India

Founder and Computational Biologist

Digirobi Solutions

<https://www.digirobi.com/>

- Developed a computational biology model for neuropathway analysis using MATLAB, leading to 1 publication on the data and enhanced understanding of molecular mechanisms.
- Analyzed cancer genomics data and developed a web-based workflow, securing 1 patent for a nutraceutical combination aimed at enhancing cancer therapies.
- Co-developed MyVivarium, an open-source cloud app for affordable, real-time research animal colony management, resulting in 1 first-author publication.
- Provided project-to-product management support and implemented business intelligence strategies to improve business performance for 3+ businesses.

06/2007 - 09/2019

India

Senior Lead Scientist

Cellworks Research India Private Limited

<https://cellworks.life/>

- Developed molecular biology and cell signaling models, including TNF, Interleukin, and metabolism pathways, improving understanding of disease mechanisms.
- Specialized in computational modeling in immunology, focusing on cell signaling and pathway models for autoimmune diseases, including rheumatoid arthritis.
- Developed a comprehensive autoimmune disease model involving 8 cell types, validated with in vivo data, contributing to 3 publications and 5 conference abstracts on immunology and skin related autoimmune diseases.
- Managed teams of 10+ members, leading projects in data analytics, computational biology, and research development to improve decision-making through advanced analytics.
- Built cancer drug/therapeutic models and developed a genomics database with 500+ cancer cell lines and TCGA and MSK datasets. This work contributed to 1 publication on leukemia genomics and 7 ASH abstracts, enhancing precision medicine by aligning drug responses with cancer genomics.
- Secured 1 approved patent on cancer drug repurposing, 2 patent applications for novel repurposed drugs for RA, and 1 process patent application on drug repurposing methodology.
- Led 25+ training sessions for team development, focusing on enhancing computational biology & data analytics in research data.

PUBLICATIONS

Intrinsic Gestational Timing Governs Human Cerebellar Development After Preterm Birth.

bioRxiv (Preprint)

September 2, 2025

Sanidas, G., Simonti, G., Ghaemmaghami, J., Woyshner, K., **Vidva, R.**, Wolff, N., Triantafyllou, M., Lowe, C., Vij, A., Sahlin Pettersen, H., Chandereng, T., Koutroulis, I., Stein-O'Brien, G., Sidiropoulos, D. N., Gallo, V. and Kratimenos, P.

[doi:10.1101/2025.09.01.673244](https://doi.org/10.1101/2025.09.01.673244)

Prematurity Reprograms Cerebellar Development and Long-Term Behavior.

bioRxiv (Preprint)

July 24, 2025

Sanidas, G., Simonti, G., Ghaemmaghami, J., Woyshner, K., Wolff, N., Byrd, C., Triantafyllou, M., Lowe, C., Salisbury, H., Goldstein, E., Sathyanesan, A., **Vidva, R.**, Koutroulis, I., Stein-O'Brien, G., Sidiropoulos, D. N., Gallo, V. and Kratimenos, P.

[doi:10.1101/2025.07.22.664624](https://doi.org/10.1101/2025.07.22.664624)

Eph/Ephrin-Mediated Immune Modulation: A Potential Therapeutic Target for Skin Cancer.

Frontiers in Immunology

April 21, 2025

Giannopoulos, K., Karikis, I., Byrd, C., Sanidas, G., Wolff, N., Triantafyllou, M., Simonti, G., **Vidva, R.**, Koutroulis, I., Theocharis, S. and Kratimenos, P.

[doi:10.3389/fimmu.2025.1539567](https://doi.org/10.3389/fimmu.2025.1539567)

MyVivarium: A cloud-based lab animal colony management application with real-time ambient sensing.

Computational and Structural Biotechnology Journal

Feb 1, 2025

Vidva, R., Raza, M.A., Prabhakaran, J., Sheikh, A., Sharp, A., Ott, H., Moore, A., Fleisher, C., Netherton, H., Goldstein, E. and Pitychoutis, P.M.

[doi:10.1016/j.csbj.2025.01.025](https://doi.org/10.1016/j.csbj.2025.01.025)

Computational analysis of cortical neuronal excitotoxicity in a large animal model of neonatal brain injury.

Journal of Neurodevelopmental Disorders

Mar 29, 2022

Kratimenos, P., Vij, A., **Vidva, R.**, Koutroulis, I., Delivoria-Papadopoulos, M., Gallo, V., & Sathyanesan, A.

[doi:10.1186/s11689-022-09431-3](https://doi.org/10.1186/s11689-022-09431-3)

Computational Models Accurately Predict Multi-Cell Biomarker Profiles in Inflammation and Cancer.

Scientific Reports

Jul 26, 2019

Fischer, C. L., Bates, A. M., Lanzel, E. A., Guthmiller, J. M., Johnson, G. K., Singh, N. K., Kumar, A., **Vidva, R.**, Abbasi, T., Vali, S., Xie, X. J., Zeng, E., & Brogden, K. A.

[doi:10.1038/s41598-019-47381-4](https://doi.org/10.1038/s41598-019-47381-4)

PUBLICATIONS

Predicting response to BET inhibitors using computational modeling: A BEAT AML project study.

Leukemia Research

Jan 7, 2019

Drusbosky, L. M., **Vidva, R.**, Gera, S., Lakshminarayana, A. V., Shyamasundar, V. P., Agrawal, A. K., Talawdekar, A., Abbasi, T., Vali, S., Tognon, C. E., Kurtz, S. E., Tyner, J. W., McWeeney, S. K., Druker, B. J., & Cogle, C. R.

[doi:10.1016/j.leukres.2018.11.010](https://doi.org/10.1016/j.leukres.2018.11.010)

Histatin 5 binds to Porphyromonas gingivalis hemagglutinin B (HagB) and alters HagB-induced chemokine responses.

Scientific Reports

Jan 29, 2014

Borgwardt, D. S., Martin, A. D., Van Hemert, J. R., Yang, J., Fischer, C. L., Recker, E. N., Nair, P. R., **Vidva, R.**, Chandrashekaraiah, S., Progulske-Fox, A., Drake, D., Cavanaugh, J. E., Vali, S., Zhang, Y., & Brogden, K. A.

[doi:10.1038/srep03904](https://doi.org/10.1038/srep03904)

Defensin DEFB103 bidirectionally regulates chemokine and cytokine responses to a pro-inflammatory stimulus.

Scientific Reports

Feb 6, 2013

Harvey, L. E., Kohlgraf, K. G., Mehalick, L. A., Raina, M., Recker, E. N., Radhakrishnan, S., Prasad, S. A., **Vidva, R.**, Progulske-Fox, A., Cavanaugh, J. E., Vali, S., & Brogden, K. A.

[doi:10.1038/srep01232](https://doi.org/10.1038/srep01232)

ABSTRACTS & CONFERENCE PRESENTATIONS

Necrotizing Enterocolitis Selectively Disrupts Pontine Development in the Preterm Brain.

Neuroscience / SfN 2025 #PSTR103.21

Nov. 16, 2025

Sanidas, G., Lowe, C., Byrd, C., **Vidva, R.**, Simonti, G., Chandereng, T., Gallo, V., & Kratimenos, P.

<https://www.abstractsonline.com/pp8/#!/21171/presentation/30651>

Early postnatal rehabilitation mitigates motor, growth, and social deficits in perinatal brain injury.

Neuroscience / SfN 2025 #PSTR019.10

Nov 15, 2025

Simonti, G., **Vidva, R.**, Sanidas, G., Byrd, C., Lowe, C., Triantafyllou, M., Wolff, N., Chandereng, T., Koutroulis, I., Gallo, V., & Kratimenos, P.

<https://www.abstractsonline.com/pp8/#!/21171/presentation/31850>

AraC-Daunorubicin-Etoposide (ADE) Response Prediction in Pediatric AML Patients Using a Computational Biology Modeling (CBM) Based Precision Medicine Workflow.

Blood / ASH 2018 #4034

Nov 29, 2018

Cogle, C. R., Abbasi, T., Singh, N. K., Sauban, M., Raman, R. K., **Vidva, R.**, Tyagi, A., Talawdekar, A. A., Das, A., Kornblau, S. M., Hoff, F. W., Horton, T. M., & Vali, S.

[doi:10.1182/blood-2018-99-115775](https://doi.org/10.1182/blood-2018-99-115775)

ABSTRACTS & CONFERENCE PRESENTATIONS

Predicting Response to Dasatinib Using a Computational Model and Its Validation: A Beat AML Project Study.

Blood / ASH 2018 #1541

Nov 29, 2018

Tyner, J. W., Druker, B. J., Tognon, C. E., Kurtz, S. E., Drusbosky, L. M., Sahu, D., **Vidva, R.**, Kapoor, S., Azam, H., Kumar, R., Chickdipatti, P. G., Raveendaran, N. T., Gopi, R. P., Abbasi, T., Vali, S., & Cogle, C. R.

[doi:10.1182/blood-2018-99-115678](https://doi.org/10.1182/blood-2018-99-115678)

Predicting Response to BET Inhibitor in Combination with Palbociclib/Sorafenib Using a Computational Model and Its Validation: A Beat AML Project Study.

Blood / ASH 2018 #1540

Nov 29, 2018

Tyner, J. W., Druker, B. J., Kurtz, S. E., Tognon, C. E., Drusbosky, L. M., **Vidva, R.**, Narvekar, Y., Agrawal, A. K., Gera, S., Prasad, S. A., Shyamasundar, V. P., Tunwer, G. S., Abbasi, T., Vali, S., & Cogle, C. R.

[doi:10.1182/blood-2018-99-115284](https://doi.org/10.1182/blood-2018-99-115284)

Predicting Response to CDK4/6 Inhibitors and Combinations Using a Computational Biology Model and Its Validation: A Beat AML Project Study.

Blood / ASH 2017 #3909

Dec 7, 2017

Drusbosky, L. M., Turcotte, M., Castillo, P., **Vidva, R.**, Gera, S., Sauban, M., Bhargav, P., Deep, P., Kapoor, S., Abbasi, T., Vali, S., Tognon, C. E., Kurtz, S. E., Tyner, J. W., Druker, B. J., & Cogle, C. R.

<https://www.sciencedirect.com/science/article/pii/S0006497119844257>

Predicting Response to BET Inhibitors Using a Computational Model and Its Validation: A Beat AML Project Study.

Blood / ASH 2017 #3908

Dec 7, 2017

Drusbosky, L. M., **Vidva, R.**, Gera, S., Lakshminarayana, A. V., Shyamasundar, V. P., Agrawal, A. K., Talawdekar, A., Abbasi, T., Vali, S., Tognon, C. E., Kurtz, S. E., Tyner, J. W., Druker, B. J., & Cogle, C. R.

<https://www.sciencedirect.com/science/article/pii/S0006497119844245>

Predicting Response to IDH1/IDH2 Inhibitors Beyond IDH Mutations Using a Computational Model and Its Validation: A Beat AML Project Study.

Blood / ASH 2017 #3907

Dec 7, 2017

Drusbosky, L. M., **Vidva, R.**, Gera, S., Das, A., Tiwari, K. K., Joseph, V., Mohapatra, S., Lunkad, N., Abbasi, T., Vali, S., Tognon, C. E., Kurtz, S. E., Tyner, J. W., Druker, B. J., & Cogle, C. R.

<https://www.sciencedirect.com/science/article/pii/S0006497119844233>

A Genomic Signature Predicting Venetoclax Treatment Response in AML Identified By Protein Network Mapping and Validated By Ex Vivo Drug Sensitivity Testing: A Beat AML Project Study.

Blood / ASH 2016 #1713

Dec 2, 2016

Drusbosky, L., Abbasi, T., Vali, S., Radhakrishnan, S., Kumar Singh, N., Usmani, S., Parashar, D., **Vidva, R.**, Druker, B. J., Tognon, C. E., Lo, P., Tyner, J. W., & Cogle, C. R.

[doi:10.1182/blood.V128.22.1713.1713](https://doi.org/10.1182/blood.V128.22.1713.1713)

ABSTRACTS & CONFERENCE PRESENTATIONS

A Predictive Model of an Oral Inflammatory Response

AADR/CADR 2014 #1369

Mar 22, 2014

Fischer, C., Abbasi, T., Vali, S., Nair, P., **Vidva, R.**, Tiwari, K., & Brogden, K. A.

<https://iadr.abstractarchives.com/abstract/43am-184905/a-predictive-model-of-an-oral-inflammatory-response>

HBD3 Both Enhances, Attenuates Dendritic Cell MAPK Responses to HagB.

IADR/AADR/CADR 2013 #2820

Mar 23, 2013

Harvey, L., Recker, E., Raina, M., Radhakrishnan, S., Prasad, S., **Vidva, R.**, Cavanaugh, J., Vali, S., & Brogden, K. A.

<https://iadr.abstractarchives.com/abstract/13iags-173181/hbd3-both-enhances-attenuates-dendritic-cell-mapk-responses-to-hagb>

Predictive Software-Based Mathematical Modeling: A Novel Approach to Development of Oral Therapies for Rheumatoid Arthritis – Validation in a Murine Collagen Induced Arthritis Model

ACR/ARHP 2012 #345

Nov 12, 2012

Singh, G., **Vidva, R.**, Nair, P., Radhakrishnan, S., Fernandes, P., Abbasi, T., Refino, C., Cruz, J. D., & Vali, S.

<https://acrabstracts.org/abstract/predictive-software-based-mathematical-modeling-a-novel-approach-to-development-of-oral-therapies-for-rheumatoid-arthritis-validation-in-a-murine-collagen-induced-arthritis-model/>

Predictive Software-Based Mathematical Modeling: A Novel Approach to Development of Oral Therapies for Rheumatoid Arthritis – Validation in a Murine Collagen Induced Arthritis Model.

ACR/ARHP 2012 #342

Nov 12, 2012

Vali, S., Refino, C., Cruz, J. D., **Vidva, R.**, Nair, P., Radhakrishnan, S., Fernandes, P., Abbasi, T., & Singh, G.

<https://acrabstracts.org/abstract/novel-combination-therapy-of-existing-repurposed-therapies-designed-by-predictive-software-modeling-shows-profound-impact-on-disease-progression-in-a-murine-collagen-induced-arthritis-model/>

Insights into the Anti-inflammatory Action of the Soluble Epoxide Hydrolase Inhibitor through Systems Biology based in silico Modeling Approach.

The FASEB Journal - Pharmacology/Experimental Therapeutics

Mar 1, 2008

Vali, S., Hegedus, C., Bhattacharjee, C., **Vidva, R.**, Schmelzer, K., Inceoglu, B., & Hammock, B. D.

[doi:10.1096/fasebj.22.1_supplement.479.13](https://doi.org/10.1096/fasebj.22.1_supplement.479.13)

PATENTS

Use of scientifically matched plant supplements combined with antineoplastic compounds for the treatment of hematological malignancies.

Patent No. US11478444B2 [**Granted**] - Oct 25, 2022

Vali, S., Abbasi, T., Stopka, T., Minarik, L., Singh, N. K., Usmani, S., Radhakrishnan, S., Sikora, H., **Vidva, R.**, & Pimkova, K.

<https://patents.google.com/patent/US11478444B2/>

Combination of nelfinavir, metformin and rosuvastatin for treating cancer caused by aberrations in PTEN/TP53.

Patent No. US10098880B2 [**Granted**] - Oct 16, 2018

Vali, S., Usmani, S., Sultana, Z., Kumar, A., Abbasi, T., & **Vidva, R.**

<https://patents.google.com/patent/US10098880B2/>

PATENTS

System and method for development of therapeutic solutions.

Patent No. US20150302167A1 [Published] - Oct 22, 2015

Vali, S., Abbasi, T., Fernandes, P., Rajagopalan, S., Alam, A., **Vidva, R.**, Nair, P. R., Kapoor, S., Radhakrishnan, S., Sultana, Z., Ramanujan, K. S., Tiwari, K. K., Kumar, A., Singh, N. K., Agrawal, A. K., Talawdekar, A. A., Usmani, S., & Singh, R.

<https://patents.google.com/patent/US20150302167A1/>

Compositions, process of preparation of said compositions and method of treating inflammatory diseases.

Patent No. US20150098993A1 [Published] - Apr 9, 2015

Vali, S., **Vidva, R.**, Nair, P. R., Fernandes, P., Abbasi, T., & Radhakrishnan, S.

<https://patents.google.com/patent/US20150098993A1/>

Compositions, process of preparation of said compositions and method of treating inflammatory diseases.

Patent No. US20140127295A1 [Published] - May 8, 2014

Vali, S., **Vidva, R.**, Nair, P. R., Fernandes, P., Abbasi, T., & Radhakrishnan, S.

<https://patents.google.com/patent/US20140127295A1/>

SKILLS

Science

Computational Biology, Bioinformatics, Mathematical Modeling, Predictive Modeling, Systems Biology, Statistical Analysis, Machine Learning, Data Mining, Data Analysis, Data Cleaning, Data Visualization, Immunology, Cancer Biology, Immuno-Oncology, Molecular Biology, Cell Biology, Cell Signaling, Genetics, Genomics, Precision Medicine, Drug Discovery, Cancer Therapeutics, Developmental Neuroscience, Nutritional Genomics, Clinical Study Design, Patent Analysis, Behavior and Locomotor Analysis.

Programming & Tools

Python, R, SQL, MATLAB, C, C++, Unix Shell, HTML/CSS/JS, PHP, Git/GitHub, Tableau, AWS, Server Administration

Management

Project Management, Project Quality Control, Leadership, Data-Driven Decision-making, Storytelling